

1. Introduction & Problem Framing

The Train ACV (Air Conditioning and Ventilation) subsystem challenge requires identifying and localising a refrigerant leakage fault to a specific car on an 8-car train. Because refrigerant leakage affects heat transport, it typically manifests as an inability for the ACV to reach or maintain target temperatures.

Unlike standard binary classification, the hackathon evaluates submissions using a **linear rank-decay score**, rewarding models that accurately rank cars from most to least likely to be faulty. To solve this, we formulated the problem as a probabilistic ranking task using a Random Forest Classifier.

2. Data Strategy & Preprocessing

A key challenge identified in the Info Kit was that parameter sets and column counts vary between case files. To make the pipeline robust against schema changes in the held-out test data:

- **Dynamic Column Extraction:** Our script dynamically groups columns by car identifier (e.g., Car 01, Car 02) rather than hardcoding column indexes.
- **Feature Standardization:** We stripped the specific Car XX - prefixes from the column names during processing. This allowed us to stack all cars into a single dataset, forcing the model to learn the fundamental physical signature of a leak rather than memorizing which car number is faulty.
- **Missing Feature Imputation:** Since the test set might lack parameters present in the training set (or vice versa), the inference pipeline checks the expected training features and safely fills missing columns with zeros, ensuring the model never crashes during inference.

3. Feature Engineering

Because the raw data is continuous multivariate telemetry sampled every 30 seconds, feeding raw time-series points directly into a standard classifier is inefficient. Instead, we transformed the temporal data into cross-sectional summary statistics for each car:

- **Statistical Aggregations:** For every numeric column (e.g., temperatures, modes), we calculated the Mean, Standard Deviation, Max, and Min. The Standard Deviation is particularly useful for identifying cars that are struggling to stabilize their temperatures.
- **Domain-Specific Feature (Temp_Diff_mean):** We engineered a synthetic feature representing the mean difference between the Indoor Average Temperature and the ACV Control Temperature (Cooling). In a healthy car, the indoor temperature should

closely track the control temperature. A refrigerant undercharge (leak) severely limits cooling capacity, causing the actual indoor temperature to diverge significantly from the target. This feature proved to be one of the highest-importance predictors in our model.

4. Model Selection & Ranking Optimization

- **Algorithm Choice:** We utilized a **Random Forest Classifier**. Random Forests are highly robust to non-linear relationships in thermodynamic data, resistant to overfitting on small datasets (we only had 6 training cases), and do not require extensive feature scaling.
- **Aligning with the Linear Rank-Decay Metric:** To maximize the hackathon's specific scoring metric, we did not use standard hard classification (`predict()`). Instead, we utilized the model's `predict_proba()` function.
- **Ranking Logic:** By extracting the probability of the "fault" class for each of the 8 cars, we sorted the cars in descending order of their fault probability. This guarantees that even if the model's confidence is low across the board, the most anomalous car is pushed to rank 1, optimizing our chances of securing maximum partial credit under the rank-decay scoring formula.

5. Ease of Use (The Application)

To ensure the solution is accessible to non-technical maintenance crew members, the model is wrapped in a lightweight Streamlit web application. Users simply upload the raw .xlsx telemetry file. The app automatically handles feature extraction, aligns the schema, computes the probabilities, and displays a clear, ranked list of cars on the screen alongside a 1-click button to download the `acv_predictions.csv` result.